Comparison Between Nesnas–Saanouni Damage-Based Cycle Jumping and the Present Chaboche UMAT Cycle-Jump Workflow

Current status: The cycle-jump thesis section is complete through Stage 8 and ready for supervisor review.

The present work is inspired by the adaptive cycle-jump idea of Nesnas–Saanouni, but it is not a direct implementation of their coupled damage-viscoplasticity formulation. The main difference is that the research paper controls the jump size using a damage variable, whereas this project adapts the same idea to a Chaboche unified viscoplastic UMAT without damage. Therefore, the damage variable is replaced by generalized Chaboche control variables such as backstress, viscoplastic strain tensor components, and residual stress.

Aspect	Nesnas–Saanouni Paper	This Chaboche UMAT Project	Main Difference
Material model	Coupled viscoplasticity-damage model	Chaboche unified viscoplastic UMAT	The present model does not include explicit damage.
Main cycle-jump control variable	Damage variable D	Generalized Chaboche control variables Y_i	D is replaced by backstress, viscoplastic strain tensor, and stress-based quantities.
Physical meaning of jump control	Limit the admissible damage growth during a jump	Limit the admissible change of selected Chaboche state variables	The jump is controlled by internal-state consistency, not damage evolution.
Admissible jump budget	Damage increment limit, often written as ΔL or admissible damage change	Admissible state change $A_i = \tau_i S_i$	The damage budget is generalized into a variable-wise state-change budget.
Jump-size formula	Based on damage evolution rate per cycle	$\Delta N_i = \left\lfloor \dfrac{\eta A_i}{\lvert \mathrm{mean}(\Delta Y_i) \rvert + \varepsilon} \right\rfloor$	Same adaptive logic, but applied to Chaboche variables instead of damage.
Global jump decision	Controlled mainly by damage evolution	Grouped minimum: $\Delta N_{\mathrm{restart}} = \min(\Delta N_X, \Delta N_{\varepsilon^{vp}}, \Delta N_S)$	The present method uses grouped controllers rather than one damage controller.
Role of accumulated plastic/viscoplastic strain	Often coupled with damage evolution	`STATEV1 = p` is used only as an accuracy monitor	Cumulative p was too conservative for jump control because it gave $\Delta N = 1$.
Restart implementation	Research-level cycle-jump framework for damage-viscoplasticity	Abaqus continuation using `SDVINI` and `SIGINI`	This project demonstrates actual Abaqus state injection and continuation.
Validation style	Paper validates the proposed damage-based cycle-jump method	Stage 5B, Stage 6D, and Stage 7C are validated against no-skip Abaqus references	Validation is focused on Abaqus UMAT workflow correctness and prediction error.
Best result in this project	Not applicable	Stage 7C: $\Delta N = 17$, 16 skipped cycles, `STATEV1` error `0.0231584782019%`, `S11` error `2.36494669088%`	The adaptive jump was directly validated in Abaqus.
Main limitation	Accuracy depends on damage-variable evolution and constitutive model assumptions	Stress/backstress extrapolation limits larger jumps	Accumulated viscoplastic strain prediction is accurate, but stress prediction becomes the limiting factor.
Scope	Full damage-viscoplastic cycle-jump method	Demonstration of Chaboche UMAT cycle-jump continuation	The present work is a workflow and methodology demonstration, not yet a full fatigue-life/damage model.

In summary, the research paper uses a damage-based adaptive cycle-jump technique, while this project generalizes the same jump-control philosophy to a Chaboche viscoplastic UMAT without damage. The main contribution of this project is the practical Abaqus implementation: predicted `STATEV` values are injected using `SDVINI`, residual stress is injected using `SIGINI`, and the continuation response is validated against no-skip Abaqus reference simulations.